디지털 논리

Boolean Algebra (II)

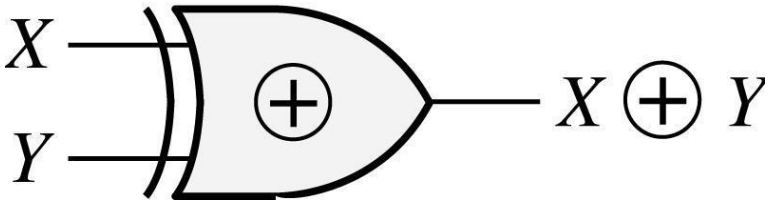
이상화

Exclusive OR (XOR) (1)

Exclusive-OR

X Y	$X \oplus Y$
0 0	0
0 1	1
1 0	1
1 1	0

Symbol



XOR gate expression

$$X \oplus Y = X'Y + XY'$$

Exclusive OR (XOR) (2)

Theorems for Exclusive-OR

$$X \oplus Y = X'Y + XY'$$

$$X \oplus 0 = X \qquad X \oplus 1 = X'$$

$$X \oplus X = 0 \qquad X \oplus X' = 1$$

$$X \oplus Y = Y \oplus X \text{ (commutative law)}$$

$$(X \oplus Y) \oplus Z = X \oplus (Y \oplus Z) = X \oplus Y \oplus Z \text{ (associative law)}$$

$$X(Y \oplus Z) = XY \oplus XZ \text{ (distributive law)}$$

$$(X \oplus Y)' = X \oplus Y' = X' \oplus Y = XY + X'Y' \text{ (Exclusive NOR)}$$

Exclusive OR (XOR) (3)

Proof by Truth table

$$X \oplus Y = X'Y + XY'$$

$$(X \oplus Y) \oplus Z = X \oplus (Y \oplus Z) = X \oplus Y \oplus Z \text{ (associative law)}$$

X	Y	Z	X⊕Y	(X⊕Y) ⊕Z	Y⊕Z	X⊕(Y⊕Z)
0	0	0	0	**0**	0	**0**
0	0	1	0	**1**	1	**1**
0	1	0	1	**1**	1	**1**
0	1	1	1	**0**	0	**0**
1	0	0	1	**1**	0	**1**
1	0	1	1	**0**	1	**0**
1	1	0	0	**0**	1	**0**
1	1	1	0	**1**	0	**1**

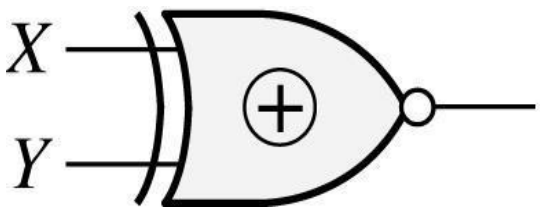
XOR = odd number of 1’ in the inputs

Exclusive NOR (1)

Equivalence operation (Exclusive-NOR)

XY	$X \equiv Y$
0 0	1
0 1	0
1 0	0
1 1	1

Symbol



$$(X \oplus Y)' = (X \equiv Y)$$

XOR gate expression

$$\boxed{X \equiv Y = XY + X'Y'}$$

Exclusive NOR (2)

Example of X-NOR

$$\boxed{F = (A'B \equiv C) + (B \oplus AC')}$$

$$F = [(A'B)C + (A'B)'C'] + [B'(AC') + B(AC')']$$
$$= A'BC + (A + B')C' + AB'C' + B(A' + C)$$
$$= B(A'C + A' + C) + C'(A + B' + AB') = B(A' + C) + C'(A + B')$$

Useful theorem:

$$\boxed{(XY' + X'Y)' = XY + X'Y'}$$

$$A' \oplus B \oplus C = [A'B' + (A')'B] \oplus C$$
$$= (A'B' + AB)C' + (A'B' + AB)'C$$
$$= (A'B' + AB)C' + (A'B + AB')C$$
$$= A'B'C' + ABC' + A'BC + AB'C$$

Consensus Theorem (1)

Consensus Theorem

$$\boxed{XY + X'Z + YZ = XY + X'Z}$$

Proof

$$\begin{aligned} XY + X'Z + YZ &= XY + X'Z + \boxed{(X + X')}YZ \\ &= (XY + XYZ) + (X'Z + X'YZ) \\ &= XY(1+Z) + X'Z(1+Y) = XY + X'Z \end{aligned}$$

Example

consensus

$$a'b' + ac + bc' + b'c + ab = a'b' + ac + bc'$$

consensus

Consensus Theorem (2)

Examples

$$A'C'D + A'BD + \cancel{BCD} + ABC + ACD'$$

$$A'C'D + \cancel{A'BD} + BCD + \cancel{ABC} + ACD'$$

Example: Reducing an expression by adding a term and eliminate.

$$F = ABCD + B'CDE + A'B' + BCE'$$

(1)

(2)

$$F = ABCD + B'CDE + A'B' + BCE' + \boxed{ACDE}$$

(3)

Consensus Term added

$$F = A'B' + BCE' + ACDE$$

Consensus Theorem (3)

Theorem for multiplying out

$$(X+Y)(X'+Z) = XZ + X'Y$$

Proof

$$(X+Y)(X'+Z) = XX' + X'Y + XZ + YZ$$

$$= X'Y + XZ + YZ = X'Y + XZ$$

consensus

Duality of consensus theorem

$$XY + X'Z + YZ = XY + X'Z$$

$$(X+Y)(X'+Z)(Y+Z) = (X+Y)(X'+Z)$$

Example

$$(a+b+c')(a+b+d')(b+c+d') = (a+b+c')(b+c+d')$$

Examples of Manipulations (1)

$$(X+Y)(X'+Z) = XZ + X'Y$$

$$(X+Y)(X+Z) = X+YZ$$

Theorem for multiplying out

$$(Q+AB')(C'D+Q') = QC'D+Q'AB'$$

proof by multiplying out

$$(Q+AB')(C'D+Q') = QC'D+\boxed{QQ'+AB'C'D}+AB'Q'$$

Redundant terms

Ex 2: Multiplying out

$$(A+B+C')(A+B+D)(A+B+E)(A+D'+E)(A'+C)$$

$$=(A+B+C'D)(A+B+E)[AC+A'(D'+E)]$$

$$=(A+B+C'DE)(AC+A'D'+A'E)$$

$$= AC+\cancel{ABC}+A'BD'+A'BE+A'C'DE$$

Examples of Manipulations (2)

Example of factoring (making product-of-sum form)

$$AC + A'BD' + A'BE + A'C'DE$$

$$= \underbrace{AC}_{XZ} + \underbrace{A'}_{X'}\underbrace{(BD' + BE + C'DE)}_{Y}$$

$$(X+Y)(X'+Z) = XZ + X'Y$$

$$= (A + BD' + BE + C'DE)(A' + C)$$

$$= [\underbrace{A + C'DE}_{X} + \underbrace{B}_{Y}(\underbrace{D' + E}_{Z})](A' + C)$$

$$X + YZ = (X + Y)(X + Z)$$

$$= (A + B + C'DE)(A + \cancel{C'DE} + D' + E)(A' + C)$$

$$= (A + B + C')(A + B + D)(A + B + E)(A + D' + E)(A' + C)$$

How to Prove an Equation

1. Construct a truth table and evaluate both sides
2. Manipulate one side by applying theorems until it is the same as the other side
3. Reduce both sides of the equation independently until the both sides become the same